

# Dhruv Arora

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## CAREER OBJECTIVE

Recent graduate with a double major in Computer Science and Data Science with a Certificate in Economic Analytics from University of Wisconsin–Madison (May 2026). Experienced in data science, AI/ML, and software development, with hands-on work in LLM training and full-stack systems. Seeking a full-time role in software engineering, data, or applied AI.

## SKILLS

- **Programming Languages:** Java, Python, C++, MATLAB, R
- **Web Development:** HTML, CSS, JavaScript, TypeScript, React
- **Databases:** SQL, SQLite, MySQL, JDBC
- **DevOps and Tools:** VS Code, Eclipse, IntelliJ, Git, Docker
- **Machine Learning & AI:** Spark, TensorFlow, Scala, PyTorch
- **Analytical Skills:** Problem Solving, Mathematical Reasoning, Data Analysis, Logical Reasoning
- **Soft Skills:** Focus, Communication, Collaboration, Adaptability

## EDUCATION

### B.S. in Computer Science and Data Science (Double Major), Certificate in Economic Analytics

University of Wisconsin-Madison, 2022 - 2026 | Cumulative GPA : 3.5

**Coursework** – Data Structures & Algorithms, Artificial Intelligence, Software Engineering, Database Management Systems, Data Science Modelling, Econometrics, Computer Engineering, Machine Organization, Human-Computer Interaction, Linear Algebra, Discrete Maths

## EXPERIENCE

### QUALCOMM (Industry-Sponsored Capstone Project), Madison, WI (ML Engineering)

January 2026 – May 2026

- Built an Android app (Kotlin) recommending songs via a 6-stage on-device multimodal ML pipeline fusing text (MobileBERT), vision (EfficientNet-B0, int8 quantized), and sensor embeddings, ranked against pre-computed Spotify embeddings by cosine similarity.
- Deployed TensorFlow Lite models with NNAPI delegation for Qualcomm Hexagon NPU acceleration on Snapdragon devices, eliminating server dependency; implemented fallback encoders to maintain app functionality when TFLite assets are unavailable.
- Authored a 12-class evaluation suite covering latency, accuracy, memory, and pipeline correctness; integrated OAuth-based Spotify access for offline song embedding generation.

### CENCORA (formerly AmerisourceBergen), Conshohocken, PA (Data Engineering Intern)

June 2025 – August 2025

- Automated SAP spool parsing in Python for a 146+ TB SAP implementation, standardizing inconsistent log formats (plain-text and box-style) and cutting manual effort by ~60%.
- Integrated Delta Lake ETL pipelines in Databricks to optimize processing speed and reduce storage overhead; orchestrated scheduled runs with Airflow and added SQL-based validation checks for timely, accurate ingestion.
- Built Power BI dashboards visualizing row counts and date-of-write trends across the largest archiving objects, enabling stakeholder-driven prioritization of archival workflows.

### UW-MADISON CDIS, Madison, WI (Undergraduate Research Assistant – LLM Training)

Aug 2025 – May 2026

- Verified and analyzed AI-generated web navigation tasks, tracking step sequences, parallelization patterns, and URL trajectories to assess retrieval complexity.
- Refined ill-posed prompts and produced structured JSON datasets to ensure reproducible data for LLM training and evaluation.

### WIPRO LIMITED, India (Full Stack Developer Intern)

June 2024 – August 2024

- Developed and optimized Java 8 applications (Employee Management System & eShop) with JDBC/MySQL, improving performance, data reliability, and CRUD efficiency by 28%.
- Implemented DAO patterns and secure database interactions using prepared statements and properties files, enhancing maintainability and scalability.

### PROLOGIC FIRST INDIA PVT. LTD., India (Software Development Intern)

June 2023 – July 2023

- Developed a Python chatbot integrating Wit.ai, Snips.ai, and Zapier, improving accuracy by 23% & boosting user engagement.
- Enhanced accessibility with speech-to-text conversion via MediaRecorder API, expanding usability for diverse users.

### UW-MADISON CDIS, Madison, WI (Peer Mentor – AI and DBMS)

January 2025 – May 2026

- Mentored 600+ students in AI & DBMS courses, covering search, ML basics, SQL, schema design, and query optimization.

### INFORMATICS SKUNKWORKS, Madison, WI (Undergraduate Student Researcher)

Jan 2024 – May 2024

- Worked under Prof. Dane Morgan to build and evaluate machine learning models using MAST-ML and scikit-learn for engineering datasets, applying cross-validation and hyperparameter tuning, which improved model accuracy and reliability.
- Delivered weekly reports on model improvement, ethical data cleaning, and reproducibility using Python and Google Collab, enhancing team understanding and ensuring data integrity.

## PROJECTS

- **Puzzle Lab - Crossword Puzzle Web App (Full-Stack Developer)** – Built using React (Next.js, TypeScript) and Quarkus (Java) with JWT authentication, REST APIs, and MySQL backend. Enabled users to create, edit, and play puzzles with personal accounts.
- **LLM Web Task Evaluation Framework (Machine Learning Engineer)** – Built a Python-based framework to benchmark LLMs on web tasks, evaluating accuracy, robustness, and failure modes via prompt-driven pipelines.
- **NOMinate - Smart Meal Planning App (Full-Stack Developer)** – Developed an intelligent meal recommendation app using OpenAI API, Python, and React, leveraging user preferences and real-time feedback to deliver personalized, non-repetitive suggestions.
- **NYC Taxi Analysis (Data Engineer)** – Developed a Databricks-based ETL and analytics pipeline for NYC Taxi data, leveraging PySpark, Delta Lake, and MLlib for predictive modelling and visualization.
- **RL - Powered Trading Strategy Optimizer (Machine Learning Engineer)** – Created a DQN-based RL agent to optimize trading strategies, boosting simulated returns by 25% and improving accuracy by 30% using Yahoo Finance data.

## EXTRACURRICULAR ACTIVITIES

- Active member of DotData Club and Software Development Club at UW–Madison, competed in Kaggle competitions and university hackathons, applying AI/ML and software development skills through coding challenges and technical workshops.